Potato - Late blight

RAG record ADV-005 | canonical key: potato_late_blight

Record ID	ADV-005
Crop / crop key	Potato / potato
Condition / disease key	Late blight / late_blight
Main pathogen	Phytophthora infestans
RAG aliases	potato late blight, Phytophthora infestans, water-soaked lesions, white sporulation

Farmer quick guide: treatments protect healthy plant tissue and slow disease spread. Dead or badly damaged leaf tissue will not become green again.

How to recognize it

- Look for pale green, water-soaked leaf spots that quickly turn brown to black.
- In cool humid weather, white fungal growth may be visible on the underside around lesion edges.
- Stems can develop dark lesions and infected tubers may show reddish-brown rot under the skin.

Weather or field conditions that favour it

Cool, humid, wet weather can make late blight spread very quickly. Infected seed tubers and cull piles can start outbreaks.

What to do now

- Do not wait for severe symptoms when late-blight weather is forecast. Protect healthy foliage early.
- Use certified disease-free seed tubers and destroy infected cull piles and heavily diseased foliage.
- Use weather-based advisories such as Indo-Blightcast where available.

Chemical treatment - only after the disease is confirmed

Chemical / formulation	Rate from cited source	How to use this information	Source
Mancozeb 75% WP	0.2% (about 2 g/L water)	CPRI lists this as a preventive contact spray when conditions	[S1]

Chemical / formulation	Rate from cited source	How to use this information	Source
		become favourable.	
Propineb 70% WP or chlorothalonil	0.2%	Alternative preventive contact options listed by CPRI.	[S1]
Cymoxanil + mancozeb	0.3%	CPRI lists this systemic mixture for use when disease is first noticed.	[S1]
Dimethomorph + mancozeb	0.3%	Another onset/systemic option listed by CPRI.	[S1]

Safety and compliance: Use only a product whose current Indian label allows the crop and target disease. Follow the label rate, water volume, re-entry interval and pre-harvest interval. Wear label-required PPE. Do not mix products unless the label or a qualified extension adviser permits it.

ICAR reported that sampled Indian P. infestans populations included metalaxyl-resistant/intermediate-resistant isolates. Do not treat metalaxyl as a universally reliable stand-alone option. [S2]

Prevention and integrated management

- Use clean certified seed.
- Destroy infected culls and volunteer potatoes.
- Ensure good coverage, including lower leaves, when spraying.
- Rotate fungicide modes of action and follow local resistance-management advice.

RAG guardrail

The AI should treat the chemical table as source-grounded examples, not permission to invent a different pesticide or dose. If the retrieved source does not match the crop, disease and formulation, return non-chemical advice and request local verification.

Sources used for this record

[S1] ICAR - Central Potato Research Institute (CPRI). Potato late blight management (Indian Horticulture, 2019).
https://www.icar.gov.in/sites/default/files/2022-06/IH-November-December-2019.pdf

[S2] Indian Council of Agricultural Research (ICAR). Annual Report 2023-24 - Phytophthora infestans and fungicide sensitivity.
https://icar.gov.in/sites/default/files/2025-04/ICAR%20Annual%20Report%202023-24-english.pdf